(SDG1)

Despite significant reductions in poverty, rural and urban poverty persist in India. How can India balance economic growth with inclusive poverty eradication strategies under SDG 1.1 and 1.3?

Strategies for Inclusive Growth Aligned with SDG 1.1 and 1.3

India has made commendable progress in poverty reduction. According to the World Bank's *Poverty and Equity Brief 2025*, extreme poverty (measured at $2.15/day in PPP terms) declined from 16% in 2011–12 to 2.3% in 2022–23. Rural and urban poverty fell to 2.8% and 1.1%, respectively. However, ground-level disparities persist, particularly in vulnerable and historically marginalised regions, underscoring the need for a holistic and inclusive poverty eradication strategy.

1. Enhancing Livelihood Opportunities

- **Skill Development & Entrepreneurship**: Expand programs such as *Swavalamban*, *PMKVY*, and *NAP* to build employability and promote self-reliance, particularly among youth and women in underserved areas.
- **Agriculture & Allied Sectors**: Make agriculture more remunerative through diversification (e.g., dairy, poultry, food processing), better access to inputs, and improved market linkages.
- **Rural Infrastructure**: Invest in roads, storage, and market infrastructure through schemes like *PMGSY* to reduce rural-urban economic gaps.
- **Non-Farm Rural Employment**: Generate opportunities in sectors like food processing and services through targeted skilling (e.g., *DDU-GKY*) and credit access.

2. Targeted Social Protection (SDG 1.3)

- **Universal Insurance Coverage**: Accelerate efforts to meet IRDAI's goal of insurance for all by 2047, especially in rural areas. Ensure that health and medical insurance are affordable and accessible.
- **Direct Benefit Transfers (DBT)**: Strengthen DBT systems via Aadhaar-linked accounts for efficient and transparent subsidy delivery.
- **Employment and Pension Schemes**: Enhance schemes like *MGNREGS* and *NSAP* to ensure income security and protection against life-cycle risks.
- **Universal Social Registry**: Establish a dynamic, integrated database to identify and reach eligible beneficiaries effectively.

3. Inclusion of Marginalised and Vulnerable Groups

- Create a supportive ecosystem for communities historically excluded from development, such as tribal populations in Jharkhand and the Northeastern states.
- Tailor policies to address their specific socio-economic challenges and improve access to education, healthcare, and markets.

4. Urban Inclusion and Informal Sector Support

- Expand affordable rental housing and urban skilling programs.
- Leverage platforms like the *E-Shram* portal to improve informal workers' access to social security and entitlements.

5. Human Capital Development

- Ensure universal access to quality education and healthcare as a long-term poverty reduction strategy.
- Strengthen nutrition interventions like *POSHAN Abhiyan* and the *Mid-Day Meal Scheme* to address malnutrition and improve learning outcomes.
- Promote awareness and behavioural change related to calorie intake and obesity through mechanisms like the Sugar Board.

6. Demographic Management and Labour Policy

- Implement effective population control measures to convert the demographic dividend into a growth asset rather than a liability.
- Develop a modern, inclusive labour policy to ensure fair wages, social security, and mobility, particularly in the gig and informal economy.

7. Fiscal Policy and Redistribution

- Adopt progressive taxation to fund social investments without fiscal stress.
- Redirect subsidies toward the most vulnerable through well-targeted schemes like *PM Ujjwala* (LPG) and the

National Food Security Act (NFSA).

Inclusive Strategies for Poverty Eradication in Line with SDG 1.1 and 1.3

Progress So Far

According to the **World Bank's Poverty and Equity Brief 2025**, India has made significant strides in poverty reduction:

- **Extreme poverty** (defined at $2.15/day in PPP terms) fell from **16% in 2011–12** to **2.3% in 2022–23**.
- **Rural and urban poverty** declined to **2.8% and 1.1%**, respectively.
- Despite this, **multidimensional poverty** remains high. The **2024 Global Multidimensional Poverty Index (MPI)** by UNDP and Oxford reports **India has the highest number of people living in multidimensional poverty**, underscoring the need for **inclusive development**, not just economic growth.

Strategic Focus Areas

1. SDG 1.1: Ending Poverty in All Its Forms

a. Skill Development and Education

- Expand access to skilling programs (*PMKVY, DDU-GKY, Swavalamban*) to improve employability.
- Prioritize **youth engagement** to prevent diversion towards anti-social elements and ensure workforce readiness.
- Promote **vocational education** with stipends and mandatory training modules.

b. Rural Economic Strengthening

- Improve **agricultural productivity** and promote **allied sectors** (dairy, poultry, food processing).
- Expand **rural infrastructure** (roads, storage, markets) via schemes like *PMGSY*.
- Encourage **non-farm rural employment** through entrepreneurship support.

c. Urban Inclusion

- Strengthen urban livelihoods via **affordable rental housing**, **skilling initiatives**, and **informal sector protections**.
- Enhance platforms like the *E-Shram* portal for worker entitlement access.

2. SDG 1.3: Implementing Social Protection Systems

a. Insurance and Pension Coverage

- Accelerate IRDAI's goal of "**insurance for all by 2047**."
- Expand **health, life, and pension coverage** under schemes like *PMJAY*, *Atal Pension Yojana*, and *NSAP*.
- Ensure affordability and penetration in remote and rural areas.

b. Digital and Technological Innovations

- Leverage **CBDC (Central Bank Digital Currency)** with **programmable features** to ensure targeted subsidy transfers and minimize leakage.
- Implement a **Universal Social Registry** for dynamic identification of eligible beneficiaries.

c. Employment Guarantees and Direct Transfers

- Strengthen **MGNREGS** to provide employment during lean agricultural periods.
- Continue **Direct Benefit Transfers (DBTs)** using Aadhaar for transparency and efficiency.

Cross-Cutting Measures for Sustainable Poverty Reduction

1. Fiscal and Taxation Reforms

- Introduce **progressive taxation**, surcharges, and cess on top 1% income earners to address inequality.
- Improve public spending efficiency and redirect subsidies toward vulnerable groups (*NFSA, PM Ujjwala*).

2. Human Capital Investment

- Ensure **universal access to quality healthcare and education**.
- Implement **price controls** for essential medicines via **NPPA** and monitor essential goods through the **Essential Commodities Act**.

3. Gender and Social Equity

- Increase **Female Labour Force Participation Rate (LFPR)** through maternity benefits, childcare support, and menstrual leave.
- Eliminate **discrimination based on caste, gender, or ethnicity**. As sociologist André Béteille emphasizes, **social identity remains a major cause of poverty** in India.

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4. Regional and Sectoral Priorities

- Focus on historically marginalized regions such as the **tribal belts of Jharkhand** and **Northeast India**.
- Promote **cooperative societies** and empower **local governments** to boost rural enterprise and participation.
- Protect vulnerable sectors like **agriculture and dairy** from cheap imports and encourage **high-value exports** (e.g., textiles, fisheries).

Tackling Poverty in India: Strategies Aligned with SDG 1.1 and 1.3

Current Status

India has made commendable progress in poverty reduction:

- **NITI Aayog's National MPI 2024** reports a decline in multidimensional poverty from **25% (2015–16)** to **11.28% (2019–21)**, lifting over **24.82 crore** people out of poverty.
- **World Bank and ILO (2025)** highlight a fall in extreme poverty from **27% to 5.3%**, and an increase in **social protection coverage from 22% to 64.3%** between 2016 and 2025.

Despite these gains, **income inequality**, **urban-rural disparities**, and **persistent vulnerabilities** highlight the need for a more integrated, adaptive, and accountable approach.

Key Strategies for Achieving SDG 1.1 and 1.3

1. Employment Generation and Skill Development

- Scale up schemes such as **MGNREGA**, **DDU-GKY**, **PM Kaushal Vikas Yojana**, and **PM SVANidhi** to provide employment opportunities, especially during lean periods.
- Empower gig and informal sector workers through **upskilling and formalization** to ensure decent incomes and stable employment.
- Leverage **MSMEs**, which contribute nearly **45% of exports**, for job creation through targeted credit, policy support, and scheme integration (*e.g., Mudra Yojana*).

2. Social Protection and Financial Inclusion

- Expand and deepen **PMJDY**, **PM-KISAN**, and **National Social Assistance Programme (NSAP)** for financial security.
- Increase **pension and health insurance coverage** under schemes like **Ayushman Bharat** and **Atal Pension Yojana**, especially to protect households from health-related impoverishment.
- Encourage the use of **programmable digital tools** (e.g., **CBDC**) for targeted subsidies and to reduce leakages and corruption.

3. Affordable Housing and Basic Services

- Strengthen access to **housing, clean cooking fuel, and basic sanitation** through **PM Awas Yojana**, **Ujjwala Yojana**, and **Swachh Bharat Mission** to improve quality of life and well-being.

4. Health and Nutrition Security

- Ensure universal access to **nutrition and healthcare services** through programs like **POSHAN Abhiyaan**, **PM Garib Kalyan Anna Yojana**, and **Rashtriya Swasthya Bima Yojana**.
- Prioritize **preventive care and coverage** in rural and low-income areas.

5. Women Empowerment and Rural Development

- Promote **Self-Help Groups (SHGs)** under **DAY-NRLM**, expand access to **credit and training**, and support women's participation in the workforce through **Stand-Up India** and maternity-related provisions.

Systemic and Structural Reforms

1. Strengthening Delivery and Implementation

- Focus on **outcome-based governance**, not just scheme creation.
- Integrate **technology (AI, blockchain, GIS)** for beneficiary tracking, transparency, and real-time monitoring.

2. Migration-Aware and Urban-Centric Policy

- Design **portable social protection frameworks** for internal migrants.
- Expand services like **affordable urban housing**, **rental assistance**, and **informal worker registration**.

3. Climate-Resilient Poverty Reduction

- Recognize climate vulnerability as a poverty multiplier.
- Integrate **climate action with poverty reduction**, especially in agriculture, housing, and disaster risk mitigation.

4. Behavioural Insights for Inclusion

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- Use **nudges**, such as targeted SMS reminders, to encourage enrolment in social schemes and improve public awareness.
- Address **cybersecurity risks** to build trust in digital platforms.

5. Inclusive Data and Measurement

- Move from household expenditure surveys to **income-based assessments** to better track inequality and target policies effectively.

6. Structural Reforms in Core Sectors

- Strengthen **healthcare**, **education**, and **skilling infrastructure** with a focus on quality, access, and accountability.
- Improve **agricultural income** by:
  - Enhancing crop productivity,
  - Reducing post-harvest losses,
  - Expanding cold storage and supply chains.

(SDG 2- Zero hunger)

What policies and programs have been introduced recently to address hunger and malnutrition in India, and are they inclusive and effective?

Addressing Hunger and Malnutrition in India: Recent Programs and Policy Interventions
India continues to face a **dual burden of malnutrition**—both **undernutrition and micronutrient deficiencies**, especially among vulnerable groups such as children, pregnant and lactating women, and adolescent girls. The government has launched a series of integrated initiatives targeting **food security, maternal and child nutrition, agricultural resilience, and healthcare access**.

1. Nutrition-Specific Interventions

a. POSHAN Abhiyaan and Mission POSHAN 2.0

- A comprehensive mission focusing on improving nutritional outcomes for children, pregnant and lactating women, and adolescent girls.
- Merged **POSHAN Abhiyaan**, **ICDS**, and **Scheme for Adolescent Girls**.
- **2025 Budget Allocation**: ₹21,960 crore; benefiting ~8 crore children, 1 crore pregnant women, and 20 lakh adolescent girls.
- **Saksham Anganwadi** initiative supports infrastructure strengthening and capacity building.
- Over **6.79 crore POSHAN Pakhwada** activities conducted nationwide to promote awareness and community involvement.

b. Integrated Child Development Services (ICDS)

- One of the oldest and most expansive schemes delivering supplementary nutrition, immunization, and health checkups through Anganwadi centers.

c. PM Matru Vandana Yojana

- Provides conditional cash incentives to pregnant and lactating women to promote health-seeking behaviour and nutrition.

d. Food Fortification Initiatives

- **Rice Fortification** under the PDS and mid-day meals to address widespread micronutrient deficiencies.
- Expansion of **fortified staples** (iron, folic acid, Vitamin B12) across key food delivery programs.

2. Food Security and Hunger Alleviation

a. PM Garib Kalyan Anna Yojana (PMGKAY)

- Launched in 2020 to address pandemic-induced hunger by providing **free food grains** to over **80 crore beneficiaries** under the National Food Security Act.

b. Antyodaya Anna Yojana (AAY)

- Targets the **poorest of the poor**, providing 35 kg of food grains per household per month at subsidized rates.

c. One Nation, One Ration Card (ONORC)

- Ensures **portability of food entitlements** across states, especially benefiting migrant workers and their families.

d. Procurement and Price Stabilization

- Government procurement of food grains ensures **price support to farmers** and a **buffer stock** for welfare schemes.
- **Export restrictions** on essential crops are imposed as needed to stabilize domestic food prices.

3. Nutrition-Linked Health Interventions

a. Eat Right India Movement

- Promotes **safe, healthy, and sustainable diets**, with a focus on reducing consumption of high-fat, salt, and sugar foods.

b. Jan Aushadhi Kendras

- Provide **affordable generic medicines** to economically weaker sections, supporting better health outcomes and reduced out-of-pocket expenditures.

4. Agricultural and Allied Sector Support

a. Crop Yield and Diversification

- **PM-AASHA**, **PMFBY**, and **Per Drop More Crop** aim to improve yield, crop insurance access, and water-efficient farming.
- **National Millet Mission** promotes **nutri-cereals** as part of a climate-resilient and nutrition-rich food system.

b. Promotion of Allied Activities

- Support for **fisheries, poultry, and animal husbandry** through dedicated schemes to enhance rural incomes and diversify nutritional sources.

c. Local Food Systems

- Encouragement of **locally produced and seasonal foods** through community awareness campaigns and POSHAN Tracker data integration.

5. Women-Centric and Community-Based Programs

a. Mission Shakti

- Integrates women's safety, empowerment, and welfare, with indirect nutritional benefits through better health and financial security.

b. Community Awareness & Behavioural Change

- Behaviour change communication strategies embedded within schemes like **POSHAN Pakhwada** and **Jan Andolan** aim to transform household nutrition practices.

India's Initiatives Toward Achieving SDG-2 (Zero Hunger)

India has made significant strides in addressing hunger and malnutrition through a range of targeted policy interventions and welfare programs. These initiatives focus on enhancing food and nutrition security, especially for vulnerable groups such as women, children, and economically disadvantaged populations.

Key Initiatives and Programs

1. Saksham Anganwadi and Mission POSHAN 2.0

- An integrated nutrition support program targeting **children, adolescent girls, pregnant women, and lactating mothers**.
- Aims to reduce **stunting, wasting, and undernutrition**, while promoting **nutritional awareness** through community engagement.
- Consolidates various schemes under one umbrella, including **Supplementary Nutrition Programme (SNP)** and **POSHAN Abhiyaan**.

2. PM POSHAN Abhiyan (Mid-Day Meal Scheme)

- Provides **hot cooked meals** and **take-home rations** to school children to improve **child nutrition**, encourage school attendance, and support learning outcomes.

3. PM Garib Kalyan Anna Yojana (PMGKAY)

- Launched during the COVID-19 pandemic to provide **free 5 kg of food grains per person per month** to over **80 crore beneficiaries**.
- Played a vital role in **alleviating hunger** during economic disruptions caused by the pandemic.

4. National Food Security Act (NFSA), 2013

- Ensures **access to subsidized food grains** for nearly **two-thirds of the population** through the **Public**

Distribution System (PDS).
- Special focus on **priority households and Antyodaya beneficiaries** (poorest of the poor).

5. One Nation, One Ration Card (ONORC)
- Enables **portability of food entitlements** across states, ensuring **food security for migrant workers and their families**.

6. Rice and Food Fortification
- Nationwide implementation of **fortified rice distribution** through PDS, mid-day meals, and ICDS to address **micronutrient deficiencies**, particularly in iron and folic acid.

7. National Health Mission (NHM)
- Though primarily a health initiative, NHM plays a complementary role in addressing malnutrition through **maternal and child health services**, immunization, and health education.

Impact and Challenges
- According to **NFHS-5 (2019–21)**, India has seen modest improvements in nutrition indicators:
  - **Stunting** in children declined from 38.4% (NFHS-4) to 35.5%.
  - **Wasting** and **underweight prevalence** have also marginally decreased.
- These improvements reflect the positive effects of government schemes, especially those focused on **maternal and child nutrition**.

However, the **Global Hunger Index (2023)** ranks India **105th out of 125 countries**, highlighting persisting issues such as:
- **Child undernutrition**, particularly in rural and tribal areas.
- **Micronutrient deficiencies** due to poor dietary diversity.
- **Implementation bottlenecks** and underutilization of allocated funds in certain states.

Way Forward
1. **Data-Driven Targeting**: Utilize **real-time monitoring tools** like the **POSHAN Tracker** for improved beneficiary tracking and scheme delivery.
2. **Community Participation**: Strengthen local governance and community-led nutrition campaigns to ensure last-mile coverage and behavioural change.
3. **Integration with Health and Agriculture**:
   - Link nutrition efforts with **maternal and child health services**.
   - Promote **nutri-cereal cultivation** (e.g., under the **Millet Mission**) to diversify diets and improve food security.
4. **Focus on Migrants and Urban Poor**:
   - Expand **urban nutrition initiatives**.
   - Ensure portability and digital access to entitlements via **ONORC**.
5. **Policy Synergy and Accountability**:
   - Enhance inter-sectoral coordination between ministries.
   - Build **accountability frameworks** at the state and district levels for scheme implementation.

(SDG 3- Good health and well being)

How can India strengthen healthcare infrastructure to achieve universal health coverage while addressing NCDs and infectious diseases?

Strengthening India's Healthcare System: Key Strategies and Policy Recommendations
India's healthcare system has made significant strides in areas like digital health, insurance penetration, and infrastructure development. However, challenges remain in terms of accessibility, affordability, quality, and equitable delivery. To address these gaps, the following targeted measures are proposed:

1. Digital Transformation of Healthcare
- **Universal Digital Health Records**: Implement a **national digital health record** for every citizen to enable better diagnosis, continuity of care, and efficient resource allocation under the **Ayushman Bharat Digital Mission**.
- **Telemedicine Infrastructure**: Expand **video consultation facilities** in government hospitals, especially in rural

and underserved regions. Leverage platforms like **eSanjeevani** and strengthen digital connectivity at primary health centers.

- **Health Data Governance**: Establish robust **data privacy and interoperability frameworks** to protect patient data while enabling seamless sharing among authorized health providers.

2. Primary Healthcare and Infrastructure Development

- **Revamp Primary Health Centres (PHCs)**: Focus on **upgrading PHCs and Health & Wellness Centres (HWCs)** with skilled personnel, better diagnostic facilities, and regular medicine supplies.
- **Promote Aarogya Mandirs**: Encourage the development of **community-based micro health centers** in rural areas to ensure easy access to first-line healthcare services.
- **Improve WASH (Water, Sanitation, and Hygiene)**: Integrate WASH initiatives with health programs to prevent communicable diseases, especially in schools and slums.
- **Bed-to-Patient Ratio**: Invest in increasing **hospital beds, intensive care units, and emergency care facilities**, particularly in tier-2 and rural areas.

3. Health Financing and Insurance Reforms

- **Universal Health Coverage**: Ensure that **health insurance coverage**, such as under **PM-JAY**, is expanded to include all vulnerable groups.
- **Simplify Insurance Claims**: Streamline and digitize the insurance claim process. Establish **ombudsman frameworks** to handle grievances related to claim rejections and unethical practices by health professionals.
- **Health Spending**: Gradually raise **public health expenditure** to **2.5% of GDP**, as per the National Health Policy target, to improve infrastructure, human resources, and R&D.

4. Regulatory Oversight and Accountability

- **Enforce Quality Standards**: Develop a **national compliance framework** mandating that all hospitals—public or private—meet minimum facility, hygiene, and service quality standards.
- **Licensing and Monitoring**: Introduce **periodic performance audits** and link hospital licensing to adherence to prescribed benchmarks.

5. Nutrition and Preventive Healthcare

- **Food Safety Regulation**: Empower agencies like **FSSAI** to conduct stricter surveillance of food quality across supply chains. Educate the public on healthy dietary practices to prevent non-communicable diseases.
- **Nutritional Literacy**: Promote campaigns linking **balanced nutrition** to disease prevention, especially in schools, urban slums, and tribal belts.

"If your food is wrong, medicine is of no use. If your food is right, medicine is of no need."
— This principle underlines the centrality of nutrition to preventive healthcare.

6. Health Education and Behavioural Change

- **School-Based Health Education**: Introduce **basic health, hygiene, and first aid education** from early grades. Include mental health and nutrition awareness as part of the school curriculum.
- **Community Health Workers**: Strengthen training for ASHAs, ANMs, and Anganwadi workers to promote preventive healthcare practices at the grassroots level.

7. Learning from Global Best Practices

- **Decentralized Healthcare Delivery**: India can adopt aspects of the **British NHS model**, particularly its emphasis on **universal access, doorstep delivery of medicines**, and **focus on preventive care**.
- **Public-Private Collaboration**: Leverage the capacity of private hospitals for innovation while ensuring equitable public health outcomes through regulation and partnerships.

Sustainable Development Goals Group Discussion

(SDG 4- Quality education)

What educational reforms or schemes has the government implemented recently, and how are they influencing learning outcomes across different groups?

Transforming Education in India: Key Government Initiatives and Their Impact

India has launched several strategic initiatives to reform its education system, ensuring **universal access, equity, foundational learning**, and **21st-century skills** for learners. These efforts are aligned with the **Sustainable Development Goal 4 (SDG-4)**: *"Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all."*

1. National Education Policy (NEP) 2020

- **Reimagines the education system** with a learner-centric approach, emphasizing **experiential and skill-based learning**.
- Replaces the old 10+2 system with a **5+3+3+4 pedagogical structure**, integrating **early childhood care**, **foundational literacy**, and **vocational training**.
- Encourages **multilingual instruction**, **flexible curricula**, and **integration of technology** across all levels.

2. PM SHRI Schools (PM Schools for Rising India)

- Model government schools being developed with a focus on **smart classrooms**, **green infrastructure**, and **innovative teaching practices**.
- Aims to set benchmarks in quality education, based on the principles of **NEP 2020**.

3. Samagra Shiksha Abhiyan

- A holistic scheme integrating **elementary, secondary, and teacher education** under one umbrella.
- Focuses on **infrastructure development**, **learning enhancement**, **gender equity**, and **inclusive education**, particularly for **marginalized groups**.

4. NIPUN Bharat Mission

- National Initiative for Proficiency in Reading with Understanding and Numeracy.
- Targets **foundational literacy and numeracy** by **Grade 3** across all states by 2026-27.
- Essential for reducing **dropout rates** and improving long-term **learning outcomes**.

5. Digital Education Initiatives

a. PM e-Vidya

- Launched during the COVID-19 pandemic to promote **One Nation, One Digital Platform**.
- Integrates platforms like **DIKSHA**, **SWAYAM**, and **radio/TV-based education** for remote learners.

b. DIKSHA Platform

- Offers digital learning content mapped to school curricula in multiple languages.
- Enables teachers and students to **access e-content, assessments**, and **training modules**.

c. SWAYAM

- A Massive Open Online Course (MOOC) platform providing **free higher education courses**, helping bridge the **urban-rural learning gap**.

6. Skill Development and Vocational Education

a. Skill India Mission

- Offers **vocational training and skill certification** for youth and school dropouts.
- Enhances employability by focusing on industry-relevant skills.

b. ULLAS (Udaan for Lifelong Learning and Advancement in Society)

- Focuses on **adult education**, especially for individuals **above the age of 14**, promoting **digital literacy, financial education**, and **citizen-centric knowledge**.

c. Atal Tinkering Labs (under Atal Innovation Mission)

- Encourage innovation among school students in areas like **robotics, AI, and machine learning**.
- Builds a **STEM-focused ecosystem** at the school level.

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7. Equity and Inclusion Measures

- **Mid-Day Meal Scheme (now PM POSHAN)**: Improves **nutritional outcomes** and boosts **school enrollment and retention**, especially among disadvantaged children.
- **Special Training for Adolescent Girls**: Targets dropouts and out-of-school girls with **STEM education and vocational training**, ensuring gender equity.
- **Scholarship Programs (e.g., SHREYAS)**: Provide **financial assistance** for higher education to **SC/ST/OBC and economically weaker sections**.

8. Pedagogical and Co-Curricular Reforms

- **No Bag Day**: Encourages **physical well-being**, **creativity**, and **experiential learning** by integrating sports, arts, and life skills into the school schedule.
- **Vocational Integration**: NEP 2020 emphasizes the inclusion of **hands-on vocational education** from **Grade 6 onwards**, linking education with livelihood.

(SDG 5- Gender equality)

What cultural and structural shifts are essential to ensure gender equality and eliminate violence against women in India?

Towards Gender Equality: Required Cultural and Structural Shifts

Achieving genuine gender equality goes beyond legal provisions it requires **a cultural transformation coupled with institutional reforms**. The following cultural and structural shifts are essential to eliminate discrimination and violence against women:

I. Cultural Shifts: Challenging Norms and Changing Mindsets

1. **Challenge Patriarchy and Gender Stereotypes**
   - Deconstruct traditional roles that limit women's autonomy.
   - Encourage **shared responsibilities** in households and caregiving.
2. **Promote Consent and End Victim-Blaming**
   - Shift from a culture of silence to one of **consent and accountability**.
   - Educate youth on **respect, boundaries, and dignity** through schools and media.
3. **Normalize Women's Autonomy**
   - Promote **decision-making power** of women in families, careers, and communities.
   - Reject the narrative of motherhood as a societal obligation or penalty.
4. **Representation in Media and Pop Culture**
   - Encourage **balanced and empowering portrayals of women** in films, advertising, and digital platforms.
   - Celebrate **female achievers** across diverse fields, not just traditional roles.
5. **Educational and Early Socialization Reforms**
   - Integrate **gender sensitivity and equality modules** in school curricula.
   - Promote **co-educational spaces** that teach collaboration, not competition, between genders.

II. Structural Shifts: Building Inclusive Systems and Institutions

1. **Strengthen Legal and Administrative Frameworks**
   - Ensure **stricter enforcement** of laws related to domestic violence, sexual harassment, and trafficking.
   - Simplify access to legal aid, especially in rural and underserved areas.
2. **Access to Education and Economic Opportunities**
   - Ensure **universal enrollment and retention** of the girl child at all levels.
   - Promote **vocational training and skill-building** to enhance employability.
3. **Promote Women's Political and Corporate Representation**
   - Increase **women's participation in decision-making bodies** at local and national levels.
   - Encourage **representation in leadership roles** across public and private sectors.
4. **Support Systems for Survivors of Violence**
   - Provide **safe shelters, legal aid, mental health care, and financial rehabilitation** for survivors.
   - Establish **community-based support systems** and crisis response teams.
5. **Gender-Responsive Budgeting and Policy Design**

- Implement **gender budgeting** to ensure that women's needs are central to fiscal planning.
- Evaluate and redesign existing schemes (e.g., PM POSHAN) to ensure **resource control by women beneficiaries**.

6. **Use of Technology for Empowerment and Safety**
   - Develop and promote **tech-based safety tools**, helplines, and geo-tagging for emergency response.
   - Leverage **digital platforms for education, financial inclusion, and awareness** campaigns.
7. **Language and Perception Reforms**
   - Move away from labels like "vulnerable" to emphasize **agency and resilience**.
   - Recognize and **value all forms of women's labor**, including domestic and informal work.

(SDG 6- Clean water and sanitation)

What integrated water resource management strategies are needed to improve water access and sanitation in both urban and rural India?

Strategies to Improve Water Access and Sanitation in India

Access to clean water and proper sanitation is fundamental to public health, economic growth, and environmental sustainability. India, with its growing population and urbanisation pressures, requires a **multi-dimensional and long-term strategy** to ensure water security and universal sanitation.

I. Sustainable Water Management Strategies

1. **Greywater Utilisation and Recycling**
   - Promote **reuse of greywater** (from sinks, showers, laundry) for landscaping, toilet flushing, and agriculture.
   - Integrate **community-level treatment plants** in urban and rural settings.
2. **Sponge Cities Concept**
   - Adopt the **Sponge City** model (as seen in China) around metropolitan areas to manage urban flooding, recharge groundwater, and enhance water retention.
   - Use **permeable pavements, green roofs**, and **urban wetlands**.
3. **Water Body Interlinking and Irrigation Network Expansion**
   - Interlink major **rivers, lakes, and canals** to ensure equitable water distribution.
   - Build **efficient irrigation systems** to reduce dependence on monsoons and overextraction of groundwater.
4. **Household-Level Water Auditing**
   - Install **individual household water meters** to promote accountability and reduce wastage in urban areas.
   - Implement tiered pricing to encourage conservation.
5. **Crop Diversification and Water-Efficient Agriculture**
   - Shift away from water-intensive crops (like paddy and sugarcane) in water-scarce regions.
   - Promote **drip irrigation, rainwater harvesting**, and **millets cultivation** under schemes like PMKSY and the International Year of Millets framework.

II. Sanitation and Infrastructure Upgradation

6. **Modern, Planned Drainage Systems**
   - Upgrade and maintain **underground drainage networks** in both rural and urban areas.
   - Prevent **open defecation** and untreated waste discharge by strengthening monitoring under the **Swachh Bharat Mission 2.0**.
7. **Enhanced Storage and Infrastructure**
   - Increase investment in **multi-purpose modern dams**, reservoirs, and **check dams** to ensure year-round availability.
   - Promote **desilting and restoration** of local tanks and ponds.
8. **Protection of Natural Resources**
   - Enforce bans on **illegal sand mining** to prevent aquifer degradation and saltwater intrusion.
   - Launch afforestation drives for **groundwater recharge** and catchment area protection.

III. Governance, Policy, and Public Engagement

9. **Robust National-Level Water Policies**
    - Strengthen and replicate successful models like the **Namami Gange Mission**, **Atal Bhujal Yojana**, and **Jal Jeevan Mission (Har Ghar Jal)**.
    - Integrate water use planning into **urban and rural development policies**.
10. **International Collaboration and Technology Transfer**

- Collaborate strategically with countries like **Israel** to adopt advanced technologies such as **desalination, water recycling**, and **efficient irrigation systems**.

11. **Public Awareness and Behavioural Change**

- Conduct sustained **IEC (Information, Education & Communication)** campaigns on water conservation and hygiene.
- Engage **community leaders, school curriculums, and media** to foster a water-wise culture.

(SDG 7- Affordable and clean energy)

Can India meet its rising energy demands while transitioning equitably to renewable energy sources?

What recent measures has the Indian government taken to promote clean and affordable energy, and are they accessible to all?

India's Ambitious Green Energy Transition: Overview

Targets and Vision

- **300 GW Renewable Energy by 2030** (Currently ~241 GW installed capacity)
- **Expansion of Nuclear Power** from 5 GW to 50 GW
- **Development of Energy Storage Systems** to ensure grid stability and sustainable supply

Key Achievements So Far

1. **50% Non-Fossil Fuel Capacity Achieved (5 years ahead of target)**
2. **Rapid Renewable Capacity Addition:** Nearly 28 GW added recently, making India the 3rd largest renewable capacity increase globally after China and the USA.
3. **Mega Renewable Energy Parks:**
    - Bhadla Solar Park, Rajasthan (2.2 GW) – one of the world's largest single-site solar farms
    - Khavda Green Energy Park, Gujarat (30 GW planned solar & wind)
4. **Grid & Storage Infrastructure Improvements:** Enhanced collaboration between government and industry for smart meters, net metering, and energy storage pilot projects.
5. **Manufacturing Boost:** ₹24,000 crore for solar module manufacturing and ₹15,600 crore for battery storage under PLI scheme.
6. **National Green Hydrogen Mission:** ₹19,744 crore allocated to develop 5 million tonnes green hydrogen capacity.
7. **Government Schemes:** PM KUSUM, PM Surya Ghar Yojana, Solar Parks, and smart grid technologies implementation.

Major Challenges

- **Financing:** A $300 million funding gap to promote clean energy projects remains a bottleneck.
- **Dependence on Coal:** India remains one of the largest coal importers with coal-based generation still accounting for ~219 GW (largest share).
- **Grid Stability:** Managing demand fluctuations and integrating intermittent renewables (solar & wind) poses technical challenges.
- **Energy Storage:** Storage technologies are crucial but currently limited, affecting renewable energy utilization.
- **DISCOM Financial Health:** Despite net metering benefits, losses to power distribution companies (DISCOMs) affect sustainability of green transition.
- **Upfront Costs:** Renewable infrastructure requires high initial investment, making affordable financing and subsidies critical.

Current Energy Demand and Supply Scenario

- **Total Installed Capacity:** ~476 GW
- **Peak Demand:** ~277 GW with large fluctuations
- **Renewable Energy Capacity:** ~226.9 GW (49% of total), aiming for 500 GW by 2030
- **Power Shortages:** Reduced drastically from 4.2% (2013-14) to 0.1% (2024-25) — significant progress.

Recent & Ongoing Government Initiatives

- **PM Surya Ghar Muft Bijli:** Free solar electricity to households.
- **PM KUSUM Yojana:** Solarization of agricultural pumps and decentralized solar projects.
- **One Sun One World One Grid (OSOWOG):** International collaboration for solar energy sharing.
- **National Bioenergy Programme:** Promoting biofuels and biomass energy.
- **PM Janman Solar Electrification:** Solar power to unelectrified areas.
- **Battery Energy Storage Systems (BESS) Plants:** Enhancing grid flexibility and storage.
- **Solar Parks Scheme:** Creation of large-scale solar power generation hubs.

Way Forward

- Scale-up **energy storage solutions** to smoothen supply-demand fluctuations.
- Mobilize **innovative financing models** and public-private partnerships to fill funding gaps.
- Reduce coal dependency by accelerating **renewable capacity installation and grid modernization**.
- Strengthen **DISCOM reforms** to improve financial health and efficient energy distribution.
- Expand **green hydrogen** and **electric mobility** initiatives for decarbonization.

India's Renewable Energy Progress and Ambitions

What are Renewable Energy Resources?

Renewable energy sources naturally replenish and include solar, wind, geothermal, etc., providing a sustainable energy supply.

Capacity and Achievements

- **2015 Goal:** 175 GW renewable capacity by 2022 (actual ~120 GW achieved).
- **Current Status (Oct 2024):**
  - India leads as the world's largest renewable energy producer.
  - Renewable sources account for **46.3%** of installed capacity (203.18 GW of 452.69 GW).
- **As of June 2025:**
  - Total renewable energy capacity: **226.9 GW**
  - Nuclear capacity: **8.8 GW**
  - Combined, renewables + nuclear = 49% of total 467 GW installed power capacity.
- **Renewable Energy Share in Electricity Generation:**
  - Increased from **16.02% (2015-16)** to **22.13% (2024-25)**.
- **Per Capita Installed Renewable Capacity:**
  - Increased from **64.04 W (2014-15)** to **156.31 W (2024-25)**.

Key Government Initiatives and Policies

- **National Solar Mission:** Affordable solar energy expansion.
- **PM-KUSUM Scheme:** Solar pumps for farmers to boost sustainable agriculture.
- **National Wind-Solar Hybrid Policy:** Encourages combined wind and solar projects for better efficiency.
- **E-Truck Incentive Scheme (PLI):** Promotes electric trucks to reduce emissions in transport.
- **Asia's Largest Battery Storage Facility:** Set up in Delhi to stabilize renewable supply.
- **PM-MBY (Mukhya Mantri Bijli Yojana):** Focused on electricity access and renewables.
- **Skill Development (Kaushal Vikas Yojana) in Agriculture:** Integrates clean energy with farming skills.
- **International Electricity Sharing Agreements:** Tripartite with Nepal and Bangladesh to enhance regional power trade.
- **Solar Capacity Expansion:** Added 24 GW in FY24, pushing solar capacity near 200 GW.
- **International Cooperation:** Agreements with Italy, Spain, UAE, New Zealand for climate-resilient clean energy projects.
- **National Targets:**

- 100 GW nuclear capacity by 2037
- 500 GW renewable installed capacity by 2030
- Net-zero carbon emissions goal by 2070

(SDG 8- Decent work and economic growth)

How can India formalize its large informal sector and ensure decent, productive employment for all?

Importance of Formalizing the Informal Sector

1. **Provide Basic Social Security:** Access to pensions, provident funds, and other welfare schemes.
2. **Basic Education & Skill Development:** Equip workers with relevant skills and knowledge for better employment.
3. **Financial Support:** Access to credit, banking services, and financial literacy.
4. **Insurance & Health Coverage:** Health insurance and workplace safety protections for informal workers.

Current Challenges

- **Large Informal Workforce:** Nearly 400 million workers; informal sector contributes about 50% of India's GDP.
- **Lack of Awareness & Access:** Limited knowledge of government schemes among informal workers.
- **Inadequate Education & Skill Training:** Only 20-30% of fresh graduates employable; skill mismatch.
- **Digital Divide:** Around half the rural population lacks internet access.
- **Regulatory & Compliance Issues:** Complexities hinder small enterprises' formalization.

Case Study 1: MSME Sector

- Contributes ~30% of GDP and 45% of exports.
- 99% are unorganised, representing untapped potential.
- **Problems:**
  - Regulatory compliance burden.
  - Digital divide limits market access and efficiency.
  - Low financial literacy and awareness.
- **Solutions:**
  - Financial inclusion campaigns (Finance Ministry drive Jul-Sep).
  - Community-based education (ULLAS initiative promoting financial literacy for 15+).
  - Bridging digital divide (Govt approval of companies like SpaceX to improve rural internet).
  - Banking access (Jan Dhan Yojana, PM Svanidhi for street vendors).

Case Study 2: Education & Skill Development

- India needs 170 million jobs by 2030.
- Only 20-30% of fresh graduates are employable due to skill gaps and outdated education.
- **Problems:**
  - Skill mismatch between education and industry needs.
  - Outdated education system and low R&D focus.
  - Large number of graduates turn to gig jobs due to lack of formal employment.
- **Solutions:**
  - Employment-oriented education (Skill India Mission).
  - Promote R&D (learning from countries like Israel).
  - Industry-led skill hubs in higher education.
  - Periodic educational reforms (NEP 2020).

Government Achievements

- Social security inclusion has increased from **19% (2015)** to **64% (2025)** (ILO data).
- Initiatives like PM Kaushal Vikas Yojana, PM Vishwakarma Yojana, e-Shram portal aim to empower informal workers.
- Jan Dhan Yojana and PM Svanidhi have improved financial access for small vendors and informal entrepreneurs.

Way Forward: Recommendations

1. **Focus on Skill-Based Education:** Shift from degree-centric to skill-centric curricula.
2. **Promote Industrial Sector Growth:** Support shift from agriculture/service-heavy informal jobs to formal industrial employment.
3. **Encourage FDI:** Create flexible, attractive policies to bring foreign investment (e.g., Foxconn).
4. **Raise Awareness:** Enhance outreach about government schemes targeting informal workers.
5. **Ground-Level Welfare Policies:** Ensure free healthcare, sanitation, and drinking water access for informal workers.
6. **Leverage Digital Infrastructure:** Expand internet access to bridge rural-urban divide.

Formalizing India's Informal Sector: Unlocking the Demographic Dividend

India boasts the world's largest demographic dividend with a labor force of about **470 million**, of which nearly **400 million (90-95%) work in the informal sector**. This sector comprises agricultural laborers, construction workers, domestic helpers, street vendors, gig workers, small shop owners, and employees of micro and small enterprises (MSEs). Despite its immense contribution to the economy—providing nearly **90% of urban employment** and accounting for about half of India's GDP—the informal sector faces critical challenges:

Key Challenges of the Informal Sector

1. **Lack of Job Security:** Many workers have seasonal or irregular jobs (e.g., agricultural laborers, festival workers).
2. **No Social Security:** Absence of benefits like Provident Fund (PF), Employees' State Insurance (ESI), health insurance, or pensions leaves their future uncertain.
3. **Irregular and Low Income:** Workers suffer from income instability and no income parity.
4. **Limited Financial Access:** Many informal workers remain outside formal banking and credit systems.
5. **Low Awareness:** Poor knowledge about government schemes and formalization benefits among informal workers.
6. **Compliance Burden on Employers:** Micro and small enterprises fear penalties and high costs linked with formal labor laws and social security contributions.

Why Formalization is Crucial

- **Social Security & Benefits:** Ensures workers get PF, ESI, health coverage (e.g., PM-JAY), maternity benefits (PMMVY), and pensions (e.g., APY).
- **Financial Inclusion:** Access to banking (Jan Dhan Yojana), credit (MUDRA loans), and insurance.
- **Skill Enhancement:** Formal skill training programs (PM Kaushal Vikas Yojana, PM Vishwakarma, ULLAS) improve employability and productivity.
- **Stability & Fair Wages:** Formal jobs provide more stable incomes and regulated working conditions.

Steps to Formalize the Informal Sector

1. **Mandatory Digital Identification:** Registration on platforms like the e-Shram portal to create a unified database of informal workers.
2. **Expand Social Security Coverage:** Broaden schemes like ESI, EPF, PM-JAY, PMMVY to cover informal workers systematically.
3. **Financial Access and Support:**
    - Promote Jan Dhan Yojana and Mudra loans for working capital without collateral.
    - Provide personal loans and credit tailored for gig and informal workers.
    - Establish financial cushions and loan restructuring options during economic downturns (recession, lockdowns).
4. **Skill Development and Training:** Enhance enrollment in PM Kaushal Vikas Yojana, PM Vishwakarma, ULLAS to equip workers with relevant skills for formal employment.
5. **Employer Incentives:** Address fears of compliance burden by offering Employment Linked Incentives (ELI) and incentives for enrolling workers in EPF and ESI schemes.
6. **Leverage Digital Technologies:** Use India's advanced digital payment and welfare distribution infrastructure to spread awareness and provide formal employment opportunities (e.g., expanded MGNREGA-type schemes

for informal workers).

7. **Sectoral Focus:** Shift emphasis toward industrial sectors and formal MSMEs which absorb semi-skilled labor and provide stable employment.

Government Initiatives Already in Place

- **Atal Pension Yojana (APY):** Pension scheme targeting informal workers.
- **Namaste Scheme:** Converts sanitation workers into entrepreneurs (Sanipreneurs).
- **Ayushman Bharat:** Health insurance for informal workers and vulnerable groups.
- **PM Kaushal Vikas Yojana:** Skill development to improve productivity.
- **PM Vishwakarma Yojana:** Focuses on artisan workers and small-scale entrepreneurs.
- **G-eM (Government e-Marketplace):** Digital platform for MSMEs and small retailers.

Additional Recommendations

- **Promote Foreign Direct Investment (FDI):** Encourage companies like Foxconn with flexible policies to formalize labor-intensive industries.
- **Raise Awareness:** Conduct campaigns to inform youth and workers about government schemes like e-Shram, PM Kaushal Vikas, and PM Vishwakarma.
- **Enhance Working Conditions:** Implement ground-level policies ensuring free healthcare, sanitation, and drinking water to improve quality of life for informal workers.

India's Digital Leap Towards Inclusive and Sustainable Development

India stands at a pivotal moment in its journey to becoming a developed nation. Leveraging its extensive digital infrastructure and vibrant innovation ecosystem, India is poised to drive **inclusive and sustainable growth**, creating a more equitable and robust industrial landscape that benefits all sections of society.

Key pillars for this transformation include:

1. **Leveraging Digital Public Infrastructure:**
   Platforms like **Aadhaar**, **UPI**, **ONDC**, and **BharatNet** provide a foundation for seamless access to services, financial inclusion, and transparent governance, enabling citizens and businesses alike to participate in the digital economy.
2. **Fostering a Culture of Innovation in Education:**
   Through the burgeoning **startup ecosystem**, government initiatives, and missions like **India AI Mission**, India is nurturing creativity and problem-solving skills in youth, preparing a future-ready workforce.
3. **Promoting Inclusivity in Industry:**
   Empowering **MSMEs**, supporting **rural industries**, and accelerating **digital skilling** programs ensures that growth is widespread and benefits marginalized and underrepresented communities.
4. **Digital Infrastructure Development:**
   Investing in **cybersecurity**, **artificial intelligence**, and next-gen technologies safeguards India's digital assets and drives innovation.

Challenges and Formalization of India's Informal Sector

According to the Ministry of Statistics and Programme Implementation (MoSPI), India's informal sector contributes nearly **50% of the GDP** and employs **over 90% of the workforce**. However, it faces significant challenges:

- **Low and Irregular Income:** Workers in agriculture, construction, street vending, etc., often earn daily wages without income stability.
- **Lack of Social Security:** Absence of provident fund, pension, or leave facilities leaves workers vulnerable.
- **Deprivation of Education:** Many informal workers come from generational poverty and are forced to drop out early to support their families.
- **Limited Access to Credit:** Without collateral or credit history, informal workers are excluded from bank loans, often resorting to high-interest moneylenders, trapping them in debt.

Strategies to Formalize the Informal Sector

1. **Encourage MSME Growth:**
   MSMEs are the backbone of India's economy, contributing 30% to GDP and 45% to exports. Providing easier credit facilities and support can unlock their full potential.

2. **Awareness of Government Schemes:**
   Government programs like **PM Vishwakarma Scheme**, **Employment Incentive Scheme**, **MGNREGA**, and **PMKVY** need wider publicity through focused campaigns to reach informal workers effectively.
3. **Strengthen Skill Development:**
   Initiatives such as **Skill India Mission** and **PMKVY** can equip informal workers with relevant skills, improving their employability and productivity.
4. **Promote Digital and Financial Inclusion:**
   Linking informal workers to the **JAM Trinity** (Jan Dhan accounts, Aadhaar, and Mobile) ensures direct benefit transfers, financial literacy, and inclusion in the formal economy.

(SDG-9 Industry, Innovation, and Infrastructure)

How can India harness digital infrastructure and innovation to promote inclusive industrial growth?

India's Digital Infrastructure: Catalyzing Innovation, Growth, and Inclusivity

India's digital infrastructure has witnessed a transformative evolution in recent years, positioning the country as a global leader in digital adoption. According to ICRIER, India ranks **3rd worldwide in digitalization**, and as per the **ACI Worldwide Report 2024**, around **49% of global real-time payment transactions occur in India**—a testament to the rapid digital shift shaping the economy.

The ongoing **Fourth Industrial Revolution** demands accelerated digitalization, and India is advancing steadily by focusing on several key drivers:

1. **Expanding Reach and Availability:**
   Digital infrastructure has significantly increased **commercial connectivity** within India. Today, **95% of banking transactions** occur digitally, broadening access to financial services even in remote areas.
2. **Benefitting from Economies of Scale:**
   Technology has made **loan accessibility easier for MSMEs** and enhanced tracking of Non-Performing Assets (NPAs), enabling better financial health and credit management.
3. **Boosting Innovation and Creativity:**
   Digital infrastructure supports real-time **quality connectivity and data flow**, which lowers logistics costs and increases efficiency across industries. It has also fostered an ecosystem where technology-driven firms like **HCL** and **Foxconn** are investing heavily—such as the semiconductor plant in Uttar Pradesh—further fueling innovation.
4. **Providing Support and Guidance to Small Businesses:**
   Digital platforms facilitate easier compliance, access to government schemes, and credit availability, empowering MSMEs and startups to scale operations with confidence.
5. **Easing Workflow Management and Risk Reduction through AI:**
   Artificial Intelligence-driven research and interventions optimize operations, reduce risks, and enable industries to utilize **natural resources more efficiently**.

Broader Impacts of Digital Innovation in India

- **Governance and Ease of Doing Business:**
  Digital innovation has enhanced transparency and streamlined processes, improving governance and making India an attractive destination for investors.
- **Agriculture and Climate Resilience:**
  Digital tools provide **weather predictions and market data**, empowering agriculture-based industries with timely information to optimize yields and reduce losses.
- **Emerging Sectors:**
  India's digital advancements are even fostering **space tourism** and other cutting-edge industries, positioning the country as a rising player in futuristic sectors.

Sustainable Development Goals Group Discussion

(SDG 10- Reduced inequalities)

What programs has the government introduced to reduce inequality across caste, gender and geography and what challenges remain?

India's Initiatives to Reduce Inequality Across Caste, Gender, and Geography

India has launched a variety of targeted programs to address social and economic disparities stemming from caste, gender, and regional differences. These initiatives aim to foster inclusive growth and empower marginalized communities.

Key Programs and Their Focus Areas

1. Reducing Caste-Based Inequality

- **Pradhan Mantri Adarsh Gram Yojana (PMAGY):** Focuses on holistic development of villages with substantial SC/ST populations.
- **Reservation Policies & Scholarships:** Pre- and Post-Matric scholarships for SC, ST, and OBC students to improve educational access.
- **Venture Capital Fund for OBCs:** To support entrepreneurship among marginalized caste groups.

2. Addressing Gender Inequality

- **Pradhan Mantri Jan Dhan Yojana (PMJDY):** 50% of accounts opened by women as of 2025, promoting financial inclusion.
- **Pradhan Mantri Mudra Yojana (PMMY):** Provides collateral-free loans to marginalized groups, including women entrepreneurs.
- **Skill Development Initiatives:** Programs like Niveshak Didi, AI KIRAN, Salambini, and Start-Up India focus on empowering women with skills and entrepreneurship opportunities.
- **Ayushman Bharat & Ujjwala Yojana:** Improve health and living conditions for women, especially in rural areas.
- **Participation in STEM and Digital Programs:** Increasing women's engagement in science, technology, and digital sectors.

3. Bridging Geographical Inequality

- **Digital Infrastructure Expansion:** BharatNet and PM-WANI aim to provide internet access in rural areas, reducing the digital divide.
- **Rural Development Schemes:** MGNREGA, NRLM, and PMAY improve employment and housing in underdeveloped regions.
- **Support for MSMEs and Rural Industries:** Facilitate credit, infrastructure, and market access for small enterprises in less developed areas.
- **Cross-Border Energy Sharing and Connectivity:** Agreements with neighboring countries to improve infrastructure and energy access.

Digital and Financial Inclusion Highlights

- **UPI Transactions:** In 2024, India recorded 16 billion UPI transactions, indicating widespread adoption in both rural and urban areas.
- **Financial Literacy Campaigns:** Efforts to increase awareness and usage of banking and credit facilities among marginalized groups through Jan Dhan accounts, PM Mudra loans, and government-led drives.
- **Education and Skill Development:** National Education Policy 2020, Ullas, and NIPUN programs target quality education and skill-building to prepare youth for employment.

Challenges Hindering Equal Progress

1. **Digital Divide:** Persistent gap between urban and rural access to technology and the internet.
2. **Financial Literacy:** Low awareness about banking and credit facilities in rural and marginalized communities.
3. **Technology Gaps:** Limited access to advanced technology slows development in backward regions.
4. **Unemployment & Skill Mismatch:** Education systems often fail to equip youth with market-relevant skills, leading to underemployment.
5. **Social Exclusion:** Discrimination and bias continue to limit opportunities for marginalized castes and women.
6. **Inadequate Healthcare and Infrastructure:** Limits overall quality of life and workforce productivity.
7. **Implementation Issues:** Poor delivery and monitoring hamper the effectiveness of welfare programs.

Telegram: We Learn Here

India's Gini Index 2025 & Efforts to Reduce Inequality
India's Position:

- Ranked **4th globally in income equality** (World Bank, 2025).
- Improvement driven by targeted government schemes addressing caste, gender, and geography.

1. Government Programs Addressing Inequality
A. Caste-Based Initiatives:

- **Reservation:** Reserved seats for SC/ST/OBC at all government levels and jobs.
- **Primitive Tribal Groups (PVTGs):** Automatic inclusion in welfare schemes.
- **Legal Protections:** Manual scavenging banned; Prevention of Atrocities Act (1989) protects marginalized castes.

B. Gender-Focused Schemes:

- **Nari Shakti Vandan Adhiniyam:** Empowers women's political participation.
- **Financial Inclusion:** Stand-Up India, Mudra Yojana, Self-Help Groups to boost women entrepreneurship.
- **Women in Defence:** Induction of women officers in high military roles (e.g., Operation Sindoor).

C. Geographic Development Programs:

- **Vibrant Villages Program:** Infrastructure development for border villages.
- **Priority Sector Lending (RBI):** Banks linked to loans provided to aspirational districts.
- **India-Myanmar-Thailand Multimodal Project:** Enhances trade and connectivity in Northeast India.

2. Persistent Challenges

- **Social Issues:** Mob lynching, caste-based discrimination, and honour killings continue.
- **Development Inequality:** UNDP's Inequality Adjusted HDI (0.44) shows large disparities despite HDI gains.
- **Women Entrepreneurs:** Credit access gap; only 20% of MSMEs led by women.
- **Security Concerns:** Border infiltrations and refugee influx strain resources in border regions.

3. Key Steps and Policy Measures
A. Geography-Related Measures

- Creation of Ministry for North East Region Development.
- Special category status and grants for northeastern and hilly states (16th Finance Commission).
- Infrastructure projects like Kaladan Multimodal and Char Dham Yatra promotion.
- Hill Area Development schemes.

B. Gender Empowerment Efforts

- Health: PM Matru Vandana Yojana improves maternal health.
- Political Representation: One-third reservation for women in Panchayats, Parliament, and state assemblies.
- Education: Scholarships (PM Vidyalaxmi, Sukanya Samriddhi) for girls.
- Employment: 90% of SHGs women-led; targeted schemes like Drone Didi, LIC Sakhi, and Mudra Yojana.
- Social Support: Cash transfers (Ladli Bahin), subsidized public transport, PM Ujjwala for clean fuel access.

C. Caste Equity Initiatives

- Reservation and scholarships in education and government jobs.
- NAMASTE scheme for sanitation workers.
- PM Vishwakarma scheme for caste-based entrepreneurship.
- PM Janjati mission to increase outreach to ST communities.
- Priority Sector Lending (PSL) to minorities.

4. Challenges in Implementation

- **Exclusion Errors:** Many needy individuals miss benefits due to documentation or lack of awareness.
- **Corruption:** Fund misuse reported in programs like Jal Jeevan Mission.
- **Rural-Urban & Digital Divide:** Unequal access to infrastructure and internet.
- **Job Market Issues:** Lack of non-farm skilled jobs; only ~47% of engineers employable despite 45% STEM graduates.
- **Informal Sector Dominance:** 90% workforce informal; nepotism and caste influence persist, hindering meritocracy.

Telegram: We Learn Here

6. **Additional Caste, Gender & Geography Inequality Programs**

Aspect	Key Schemes	Challenges
Caste	Stand Up India, Pre/Post Matric Scholarships, PM Ajay	Social exclusion, poor fund allocation, limited higher education access
Gender	Beti Bachao Beti Padhao, Mahila Samman, PM Matru Vandana, Ujjwala Yojana	Gender violence, wage gap, low female labor participation
Geography	PM Kisan, Smart Cities, Digital India, Aspirational Districts	Rural migration, digital divide, corruption

(SDG 11- Sustainable Cities and Communities)

With rapid urbanization, how can Indian cities become more sustainable, inclusive, and resilient?

Sustainable Development: A Path for Indian Cities
Definition:

- As per the Brundtland Commission, sustainable development means:
  "Development that meets the needs of the present without compromising the ability of future generations to meet their own needs."

Challenges Indian Cities Face on the Path to Sustainability

1. **Increasing Pollution Levels**
   - Urban air and water pollution continue to worsen.
   - Government initiatives like **Mission LIFE** and **"Ek Ped Maa Ke Naam"** aim to restore natural carbon sinks and promote environmental awareness.
2. **Lack of Environmental Safety Standards**
   - Enforcement of environmental laws and regulations such as the **Environment Protection Act** and **CSR compliance** are inconsistent or weak.
3. **Infrastructure Development Focused on Present Needs**
   - Infrastructure often built without long-term sustainability in mind.
   - Programs like the **Smart Cities Mission** aim to develop 100 smart cities with improved infrastructure and urban mobility (e.g., metro systems) to better serve citizens now and in the future.

Inclusive Development & Livability

- According to the **Global Livability Index Report 2025**, the most livable cities worldwide are consistently the happiest and least stressful for residents.
- Examples: **Copenhagen (top ranked), Switzerland, Norway.**
- Indian metro cities like **Mumbai** and **Delhi** do **not** even rank in the top 100 globally.

Key Observations from Leading Livable Cities

- Efficient and accessible public transport systems reduce pollution and commute time.
- Reduced commute stress contributes significantly to citizens' mental well-being and overall happiness.
- A sustainable city fosters not only good infrastructure but also the health and well-being of its residents.

The Road Ahead for India

- Sustainable urban development isn't just about fixing infrastructure gaps.
- Well-planned city structures can serve as catalysts for economic growth, social equity, and environmental protection.
- This, in turn, reduces stress and enhances the overall quality of life for urban residents.

- India needs to focus on integrated urban planning that balances environmental sustainability, inclusivity, and well-being.

(SDG 12- Responsible Consumption and Production)

How can India transition to sustainable production and consumption models without compromising economic growth?

Driving India's Sustainable Growth: R&D, Partnerships, and Consumption-Based Economy

1. Investing in Research and Development (R&D)

- Currently, India spends less than **1% of its GDP** on R&D, which is low compared to global leaders.
- Increasing R&D investment is crucial to **drive innovation**, improve production efficiency, and ensure sustainable, unconstrained economic growth.
- More funding will help develop **homegrown technologies** and innovations tailored to India's unique needs.

2. Strategic Partnerships with Foreign Think Tanks and Institutes

- Collaborations with leading international institutions can accelerate knowledge transfer.
- Example: **Norman Borlaug**, whose work led to India's Green Revolution, exemplifies the impact of global cooperation.
- Partnerships, such as financial support to **CIMMYT** (International Maize and Wheat Improvement Center), can help leverage global best practices to boost India's agricultural productivity and sustainability.

3. Embracing and Educating on Best Practices

- India's diversity demands standardized, scalable **best practices** in various sectors.
- Promoting sustainability and efficiency requires **education and training** of all stakeholders — from farmers to industrial workers to urban consumers.
- This ensures long-term economic growth with minimal resource wastage.

Consumption-Based Model: Pay-As-You-Use for Sustainability

- The consumption-based model encourages users to **pay only for what they use**, promoting efficient resource utilization and spreading awareness about sustainable consumption.
- This model aligns with the **Just-In-Time (JIT) system**, which minimizes waste by using resources precisely when needed, improving cost efficiency without hindering growth.

Applying Consumption-Based Model in India

Role of Government:

1. Encourage **behavioral change** through positive reinforcement techniques.
2. Launch **awareness campaigns** to educate the public on sustainability.
3. Establish **robust feedback mechanisms** and strengthen R&D to resolve emerging issues swiftly.
4. Foster ecosystems promoting **environmental consciousness** (e.g., Mission LIFE).
5. Simplify **regulatory compliance** to empower consumers to feel in control of their usage, thereby increasing motivation towards sustainable choices.

Role of Consumers:

1. Support systems enabling purchase **based on actual current needs** rather than bulk hoarding.
    - Example: Quick-commerce companies like **Blinkit** thrive by matching the Indian consumer behavior of small, frequent purchases.
2. Promote **equitable resource distribution** to meet demands fairly across India's large population with limited natural resources.

(SDG 13- Climate Action)

How can India fulfill its net zero 2070 pledge while ensuring economic development and energy security.

India's Roadmap to Net Zero Emissions by 2070: A Sustainable and Inclusive Strategy

India's ambitious pledge to achieve **net zero emissions by 2070** aligns with its commitment to climate resilience while

ensuring **energy security** and **inclusive economic growth** amid a growing population and expanding economy.

Key Strategies for Achieving Net Zero

1. Accelerate Renewable Energy Deployment

- Expand capacity in **solar, wind, hydro, and bioenergy** to achieve **500 GW by 2030**.
- Scale up flagship schemes such as **PM Surya Ghar Muft Bijli Yojana** for free electricity in new houses.
- Utilize wastelands for large-scale solar park projects to maximize land efficiency.
- Support schemes: **PM KUSUM (solar pumps), National Electric Mobility Mission Plan (NEMMP), National Action Plan on Climate Change (NAPCC), LiFE Mission**.

2. Invest in Green Technologies and Innovation

- Promote **electric vehicles (EVs), green hydrogen, battery storage, and carbon capture technologies**.
- Support domestic manufacturing through schemes like **Faster Adoption and Manufacturing of Hybrid & Electric Vehicles (FAME)** and **PM E-Drive**.
- Provide **tax incentives** for green startups, MSMEs, and homeowners investing in green infrastructure.

3. Green Urban Planning and Biodiversity Protection

- Implement **green urban planning** principles emphasizing sustainable housing, efficient public transport, and pollution control.
- Protect critical biodiversity hotspots like **Aarey Forest (Mumbai)** and **Gachibowli Forest (Hyderabad)** to conserve carbon sinks and natural climate buffers.
- Encourage **community-led afforestation** and expand **mangrove restoration** for coastal resilience.

4. Green and Blue Finance

- Mobilize **green finance** (investment in renewable energy, energy efficiency) and **blue finance** (sustainable use of ocean resources).
- Issue **sovereign green bonds** to attract climate-conscious investors.
- Offer **tax benefits and subsidies** to accelerate green investments.

5. International Cooperation

- Lead and participate in global initiatives such as the **International Solar Alliance (ISA)** to share technology, expertise, and resources.
- Collaborate on reducing oil import dependency through diversification of energy sources and joint research.

6. Just Transition and Inclusive Growth

- Ensure energy transition includes **reskilling and upskilling** of workers from fossil fuel-based industries.
- Promote **energy access** and affordability to avoid exclusion of marginalized communities.

(SDG 14- Life Below Water)

What marine and coastal conservation strategies has the government introduced, and are they integrated into national development goals?

SDG 14: Life Below Water

Goal: Conserve and sustainably use oceans, marine resources for sustainable development by reducing pollution, preventing overfishing, and protecting marine ecosystems.

India's Marine Context

- **Shoreline:** 8,118 km
- **Continental shelf:** ~3.7 lakh sq km, spanning 9 states and 7 union territories
- **Biodiversity:** 600 protected areas across ~1.6 lakh sq km (4.9% of India's geographical area protected)

Key National Initiatives

1. Blue Revolution (Neel Kranti Mission)

- Target: Increase fish production & promote sustainable fisheries
- Scheme: **Pradhan Mantri Matsya Sampada Yojana (PMMSY)**
- Progress: 10% rise in fish production; 1000+ Fish Farmer Producer Organisations (FFPOs) formed since 2022

2. Deep Ocean Mission (DOM) / Samudrayan Project

- Target: Deep-sea exploration & sustainable resource use
- Highlight: Manned submersible **MATSYA 6000** (6000 m depth) planned for 2026 to boost ocean research

3. National Coastal Mission (NCM) / Integrated Coastal Zone Management (ICZM)

- Target: Holistic coastal conservation including mangroves, coral reefs, pollution, erosion
- Progress: Mangrove plantations, degraded ecosystem restoration, awareness drives implemented

4. Swachh Sagar, Surakshit Sagar (Clean & Safe Ocean)

- Target: Clean 1000+ km coastline; remove marine plastic waste
- Progress: 50,000+ tonnes of plastic removed; 1000 km coastline cleaned since 2022

Marine Protected Areas (MPAs)

- Only **6% of India's ocean** is under protection: 2.4% fully protected, 3.6% partially protected
- MPAs impose stricter regulation on human activities to conserve marine biodiversity

Supporting Systems

SAHAV Portal

- Launched June 2025
- Collects & analyzes coastal water quality, mangrove cover, and marine protected area data for transparent, science-based management

Policy & Scheme Highlights

- **Coastal Regulation Zone Notification (2019):** Balances development with ecological protection
- **Integrated Coastal Zone Management Programme:** Ensures sustainable coastal development
- **PM Matsya Sampada Yojana:** ₹13 crore for artificial reefs in Kerala (2024)
- **MISHTI:** Mangrove restoration (22,500+ hectares restored; Gujarat leads)
- **O-SMART:** Marine pollution monitoring, coral bleaching alerts
- **Swachh Sagar, Surakshit Sagar Campaign:** Plastic cleanup along coasts

Recent Developments (2024–2025)

- **Marine Fisheries Census 2025:** Digital geotagging of 1.2 million fishing households via VyAS-NAV app
- **Shark Conservation:** Crackdown on illegal shark fishing
- **Coral Biodiversity:** Chintapalle (Visakhapatnam) identified as coral hotspot with 15 coral species
- **Artificial Reefs:** Projects in Kerala (Thiruvananthapuram, Kollam, Ernakulam) to boost biodiversity

Challenges

- Scaling up conservation efforts to meet ambitious targets
- Balancing economic needs of coastal communities with ecosystem protection
- Combating marine pollution and overfishing persistently
- Enhancing scientific research and monitoring capabilities

(SDG 15- Life On Land)

What government-led forest, biodiversity, and wildlife conservation programs have been active recently, and how are they monitored?

Ecosystem Conservation: The Need and Government Initiatives

"If the bee disappeared off the surface of the Earth, man would have only four years to live." – Albert Einstein

Every seemingly small element of the ecosystem—whether animals, forests, or biodiversity—plays a critical role in maintaining ecological balance. To sustain life on Earth, we must take responsibility for our actions and build harmonious relationships with our environment.

Government Initiatives to Foster Ecosystems

1. Forest Protection Initiatives

- **Green India Initiative (GII) 2.0:**
  Aimed at creating tree cover to act as carbon sinks. Recent efforts include restoring the Aravali hills with a plan to plant a 5 lakh tree chain and rehabilitate degraded/vulnerable hills.
- **Environmental Protection Act:**
  Empowers the government to formulate rules and regulations promoting forest and ecosystem protection.

- **Achievements:**
  India pledged to create a carbon sink of 2.5–3 billion tons and has already achieved 2.1 billion tons.

2. Wildlife Conservation Initiatives

India leads several global and national programs to protect its rich wildlife:

- **Big Cat Alliance:**
  A coalition of 27 countries working together to conserve seven big cat species.
- **Project Elephant (Since 1992):**
  Focuses on elephant conservation and habitat management.
- **Project Tiger:**
  Has successfully doubled the tiger population since inception.
- **Project Snow Leopard:**
  Conducted its first census last year; Ladakh became the first state to implement it.
- **Lion Conservation:**
  Gir National Park has recorded the highest growth in lion population.
- **Cheetah Reintroduction Initiative:**
  Bringing back cheetahs to India to restore ecological balance.
- **India's First Wildlife Corridor:**
  Established along the Delhi-Mumbai Expressway to facilitate safe animal movement.
- **Additional efforts:**
  Hornbill conservation in Tamil Nadu, Tahr deer conservation in Jammu & Kashmir, systemic plant conservation in Madhya Pradesh.

Monitoring and Enforcement

- **Wildlife Protection Act:**
  Provides the legal framework for conservation, prevention of poaching, and promotion of wildlife welfare.
- **Collaborative Local Efforts:**
  Partnerships between government and local tribal communities foster shared responsibility for ecosystem health.
- **Ministry of Environment, Forest and Climate Change (MoEFCC):**
  Oversees policy formulation and effective implementation.
- **Technological Advances:**
  - Madhya Pradesh became the first state to use digital mapping for real-time tracking of illegal tree cutting.
  - Use of drones and AI surveillance for monitoring forests and wildlife.
- **Regular Population Surveys:**
  Ongoing monitoring of endangered species like Asiatic lions and leopards to guide conservation interventions.
- **Community Engagement:**
  Initiatives such as Amrit Sarovars engage local stakeholders to protect biodiversity.
- **Independent Audits:**
  Committees evaluate progress on major projects like the Namami Gange rejuvenation program.

Other Notable Initiatives

- **Plant a Tree in the Name of Your Mother:** Encourages community tree plantation.
- **Namami Gange and Project Yamuna:** River rejuvenation projects with ecological and social benefits.
- **Artificial Coral Reef Development:** To protect marine biodiversity.

How can India ensure transparent governance, accessible justice, and reduced corruption in public systems?

Ensuring Transparent Governance, Accessible Justice & Reduced Corruption in India (SDG 16)

Introduction:

Transparent governance, accessible justice, and corruption reduction are fundamental pillars for the growth and sustainability of any society, especially a developing country like India. However, India faces multiple challenges that hinder these objectives, including bureaucratic red-tapism, political interference, outdated systems, and weak vigilance mechanisms. To address these, a combination of digital innovation, institutional reforms, and citizen participation is essential.

Challenges:

1. **Bureaucratic Red-Tapism:**
   Delays and excessive procedural formalities stall governance and justice delivery. Many innovative ideas fail to scale due to administrative inertia.
2. **Negative Attitude & Corruption Culture:**
   A system plagued by corruption often punishes honest officials, while shortcuts are normalized by citizens and officials alike.
3. **Weak Vigilance & Political Influence:**
   Independent institutions suffer from political interference, resource shortages, and outdated monitoring tools, weakening anti-corruption efforts.
4. **Colonial Legacy:**
   Outdated bureaucratic and judicial procedures reduce efficiency and limit access to justice for common people.

Policy Measures:

1. **Digitalization for Transparency:**
   - Expand initiatives like **Digital India**, **Government e-Marketplace (GeM)**, and **DigiLocker** to digitize services, reduce human interface, and minimize corruption opportunities.
   - Implement real-time tracking tools for case management (e-Courts) and public grievance redressal systems (CPGRAMS).
2. **Strengthening Independent Institutions:**
   - Ensure **autonomy** for bodies like the **Lokpal**, **CVC**, **CAG**, and **State Lokayuktas** to investigate and prosecute corruption without political pressure.
   - Adopt bipartisan procedures for appointing key officials to safeguard independence.
3. **Judicial Reforms:**
   - Set up more **fast-track courts** and strengthen **alternate dispute resolution (ADR)** mechanisms like **lok adalats** and **tele-law** to reduce pendency and improve access.
   - Digitize court records and hearings to expedite justice delivery and improve transparency.
4. **Audit and Accountability:**
   - Conduct **regular independent audits** of government departments and schemes to detect and deter corrupt practices.
   - Establish performance review mechanisms for public officials to ensure accountability.
5. **Citizen Engagement and Transparency:**
   - Promote tools like the **Right to Information (RTI)** Act and platforms such as **MyGov** for public participation and oversight.
   - Mandate public consultations on key policies and projects to increase transparency.
6. **Police and Bureaucratic Reforms:**
   - Implement the **Model Police Act** for modern, accountable policing with citizen oversight.
   - Introduce **single-window clearance systems** at local levels to simplify governance and reduce opportunities for graft.
7. **Training and Behavioral Change:**
   - Conduct regular **training and sensitization programs** for government employees to promote ethical

conduct and awareness of anti-corruption laws.

(SDG 17- Partnership for the goals)

What recent efforts has the government made to mobilize global cooperation, funding, or knowledge sharing for SDG acceleration.

The Government of India (GoI) has actively pursued several initiatives to foster global cooperation, mobilize funding, and facilitate knowledge sharing for accelerating Sustainable Development Goals (SDGs). These efforts encompass multilateral alliances, bilateral agreements, innovative financing mechanisms, and strategic partnerships:

1. **International Solar Alliance (ISA):**
   A global coalition aimed at enhancing solar energy capacity by providing technical expertise and funding support. ISA targets raising $100 million to finance solar projects worldwide.
2. **International Big Cat Alliance (IBCA):**
   Established in 2023 and joined by over 30 countries, this alliance promotes global conservation of big cats through intergovernmental cooperation, pooling funding, and sharing expertise.
3. **Coalition for Disaster Resilient Infrastructure (CDRI):**
   Launched by India in 2019, CDRI serves as a knowledge-sharing platform to promote the design and construction of disaster-resilient infrastructure globally.
4. **Renewable Energy Investment Fund:**
   A joint $600 million fund by India and Japan supporting projects in solar energy, green hydrogen, battery manufacturing, and critical mineral processing.
5. **Green Bonds:**
   The Government plans to raise ₹48,000 crore through green bonds in FY26 to finance sustainable development projects, with ₹10,000 crore already raised in H1 FY26.
6. **SDG Localization and Monitoring Tools:**
   - *SDG Index India*: Tracks progress on SDGs at the state level.
   - *Localized SDG Index*: Measures SDG progress at the Gram Panchayat level, integrating local development needs.
7. **GOI-UN Sustainable Development Cooperation Framework (2023-2027):**
   A partnership between NITI Aayog and the UN in India to enhance SDG localization and promote South-South cooperation.
8. **UN India SDG Country Fund:**
   Launched in April 2025, led by the UN Resident Coordinator in India, with an initial $3 million seed fund from the Bill & Melinda Gates Foundation to support innovative SDG-related projects.
9. **G20 2023 SDG Action Plan and Triangular Cooperation:**
   During India's G20 presidency, the group adopted an action plan to mobilize development financing and established a Digital Public Infrastructure (DPI) repository to accelerate SDG implementation.
10. **International Commitments and Multilateral Pledges:**
    Over 50 countries endorsed the Seville Commitment to triple Multilateral Development Bank (MDB) lending, increase domestic revenue mobilization, introduce innovative taxation, and establish sustainable finance standards.
11. **Regional and Global Partnerships:**
    - *BIMSTEC*: Cooperation among Bay of Bengal nations to promote regional development and diplomatic ties.
    - *QUAD*: A strategic grouping including India, the US, Japan, and Australia focused on security and development collaboration.
    - *BRICS*: A coalition representing the Global South to advocate for equitable trade, prosperity, and growth.
    - *Bilateral Free Trade Agreements*: Agreements such as with EFTA and CETA (UK) to boost trade, reduce barriers, and attract foreign investments.
12. **International Education Collaborations:**
    Bilateral agreements with universities including Southampton, Liverpool, and the University of Chicago to establish world-class campuses in India and enhance educational standards.
13. **Bilateral Cooperation on Clean Energy:**

Partnership with France to develop Small Modular Reactors (SMRs) aimed at clean and sustainable energy generation.

14. **Bilateral Maritime Cooperation:**
    "Blue Talks" with France and Chile focus on sustainable maritime engagement and blue economy initiatives.